

Instructor Performance and Course Quality Evaluation on EduPro

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Abstract - Due to the recent explosion in the field of e-learning, there is an urgent need for a structured evaluation process of the performance of instructors and quality of courses based on data analytics. The current study designs and implements EduPro - a role-based system for e-learning analytics built using Python and Streamlit - to fulfill this need using a module of Instructor Performance and Course Quality Evaluation. The entire system works with four Excel files containing the information of 3,000 students, 60 instructors, 60 courses, and 10,000 transactions in 12 subject categories. The data pipeline architecture includes Pandas-based data input from Excel files with session caching, extraction of time periods, data merge on relational keys across multiple data sources, Plotly-based interactive graphs, and session state authorization access control. Through the quantitative analysis process, we obtained a Pearson's correlation coefficient value of $r = 0.34$ between the two variables (instructor experience and ratings). Revenue analysis showed that the Data Science, Web Development, and Machine Learning courses generated the most income. These findings indicate that it is possible to provide institutional-grade analytics within a simple file-based solution.

Keywords - Instructor Performance Evaluation; Course Quality Analysis; Learning Analytics; Educational Data Mining; Multi-Role Dashboard; Streamlit; Plotly; Online Education; Institutional Analytics; Role-Based Access Control

I. INTRODUCTION

Each student enrollment, course rating, and payment transaction on an online learning platform represents an individual data point. When combined across thousands of users, this data forms a valuable dataset that can reveal which instructors are performing well, which courses satisfy learners, and where administrative attention is needed. However, most platforms do not use this data effectively. Instructor performance, course quality, and learner behavior are often stored separately, making it difficult for administrators, instructors, and students to gain a unified analytical view. This lack of integration creates challenges at multiple levels. Administrators often rely on subjective judgment when evaluating instructors due to the absence of structured data. Instructors cannot easily compare their performance within their domain or track revenue trends over time. Students, in turn, have limited information about instructor quality before enrolling in a course. Over time, this leads to repeated uninformed decisions, unresolved performance gaps, and limited institutional improvement.

EduPro addresses this issue through a dedicated Instructor Performance and Course Quality Evaluation module. After logging in, each instructor can access a personalized KPI dashboard generated from institutional Excel data and tailored to their domain. The dashboard allows instructors to compare their experience and ratings with peers using a scatter chart, analyze domain-level averages, review course-wise enrollments and ratings, and track monthly revenue and enrollment trends. Administrators have access to a platform-wide view that includes a leaderboard ranking instructors across multiple performance metrics. The system is

implemented as a single Python file using Streamlit, eliminating the need for server setup or database configuration.

II. LITERATURE REVIEW AND GAP ANALYSIS

The field of Learning Analytics has gradually moved from theory to real-world application. George Siemens and Phil Long [1] first introduced the concept, but their work stayed mostly at a theoretical level. Later, Rebecca Ferguson [2] pointed out that real-time and easy-to-use tools for educators were still missing.

In Educational Data Mining, studies by Cristóbal Romero and Sebastián Ventura [3] explored different techniques, but most of them worked as separate analyses rather than part of a single system. Similarly, Nguyen [4] focused mainly on student results, with very little attention given to instructors. Katrien Verbert et al. [5] showed that dashboards can support decision-making, but their designs were limited to single users. Further research by Muhammad Khalil and Martin Ebner [6], and Zoi Papamitsiou and Anastasios Economides [7], discussed practical issues like privacy and system complexity. Studies [8] and [9] also stressed the need for systems that are both simple to use and scalable. A review by Aldowah et al. [10] clearly showed that instructor analytics is still not widely explored. Supporting this, studies [11], [12], and [13] suggested that combining different types of analytics would make systems more useful.

From a prediction point of view, Rastrollo-Guerrero et al. [14] found that methods like Gradient Boosting perform better than single models. Finally, Dirk Ifenthaler and Jane Yau [15] explained that combining descriptive, predictive, and prescriptive analytics gives better results.

Overall, past studies show clear gaps such as lack of real-time tools, limited system integration, and less focus on instructors. EduPro is designed to address these by providing a real-time dashboard, including instructor analytics, and supporting future improvements through a flexible system design.

TABLE I. LITERATURE SURVEY: GAPS ADDRESSED BY EDUPRO

Ref(Title)	Authors	Year	Limitation	Advantages
[1] Learning Analytics: Envisioning a Research Discipline	Siemens & Long	2011	Purely conceptual; no system	Provides practical implementation through a real-time analytics dashboard
[2] Learning Analytics: Drivers, Developments and Challenges	Ferguson	2012	No real-time analytics	Enables instant insights using live data processing
[3] Educational Data Mining: A Review of the State of Art	Romero & Ventura	2010	No integrated platform	Combines multiple analytics into a unified system
[4] Student Engagement and Performance in Online Learning	Nguyen	2015	No instructor performance view	Includes detailed instructor performance tracking
[5] Learning Dashboards: An	Verbert et al.	2013	Read-only, single-role	Supports multi role interaction

Overview				with actionable insights
[6] Learning Analytics in MOOCs: Challenges and Opportunities	Khalil & Ebner	2015	No practical implementation	Simple and deployable architecture without heavy setup
[7] Learning Analytics and Educational Data Mining in Practice	Papamitsiou & Eco.	2014	No real-world deployment	Designed for real-time, practical use in institutions
[8] Dropout Prediction in MOOCs	Xing et al.	2016	No dashboard integration	Presents results through interactive dashboard views
[9] Student Success Prediction in MOOCs	Gardner & Brooks	2018	No institutional context	Uses real institutional data for meaningful insights
[10] Educational Data Mining: A Systematic Review	Aldowah et al.	2019	No instructor analytics	Introduces instructor-based KPIs and performance ranking
[11] Early Alert Systems for Student Success	Jayaprakash et al.	2014	No instructor intervention	Connects analytics directly with instructor actions
[12] Ethics and Privacy in Learning Analytics	Slade & Prinsloo	2013	No technical implementation	Ensures secure, role-based data access
[13] Predictive Analytics in Education	Kloft et al.	2014	No dashboard integration	Supports future integration with machine learning models
[14] Prediction of Student Performance using Ensemble Methods	Rastrollo-Guerrero	2020	No multi-role layer	Combines analytics with multi-user accessibility

III. SYSTEM ARCHITECTURE

EduPro is organized as a five-layer pipeline comprising the following components: Dataset, Preprocessing, Analytics, Multi-Role Application, and Session Persistence. This layered separation was a deliberate architectural decision that allows any individual layer to be extended or replaced without introducing cascading modifications across the remaining layers.

A. Data Layer

The platform operates on four Excel worksheets. The Users sheet contains 3,000 learner records. The Teachers sheet holds 60 instructor profiles with fields for domain expertise, years of experience, and ratings. The Courses sheet lists 60 course entries across 12 academic domains, each annotated with category, difficulty level, price, and rating. The Transactions sheet constitutes the largest dataset, comprising 10,000 rows of enrollment and payment records linked by TeacherID, CourseID, and UserID. On application startup, a single `pd.ExcelFile` call loads all four worksheets. The `@st.cache_data` decorator ensures this operation executes only once per session, preventing redundant I/O on subsequent reruns.

B. Preprocessing Layer

Following data ingestion, TransactionDate strings are parsed into datetime objects using `pd.to_datetime` with error coercion. Month, Quarter, and Year columns are extracted from the resulting datetime fields for subsequent time-based aggregation. The four datasets are then integrated via sequential left joins on CourseID and TeacherID, producing a consolidated master DataFrame. Instructor-level aggregate features (TotalEnrollments, TotalRevenue, AvgCourseRating, CoursesCount) are computed through groupby

operations and merged back into the teachers DataFrame. Missing numeric values are imputed with zero.

C. Analytics and Application Layers

Analytics functionality is organized across three tiers: individual KPI metrics for instructors, course-level quality visualizations, and time-series trend charts. Upon authentication, the system routes each user to the appropriate role-specific dashboard (Student, Instructor, or Admin) through Streamlit session state management (Fig. 1). All 3,061 user accounts are programmatically generated from the Excel datasets at application startup, eliminating the need for manual configuration. A shared `PLOTLY_BASE` configuration dictionary enforces consistent dark-theme styling across all rendered Plotly charts.

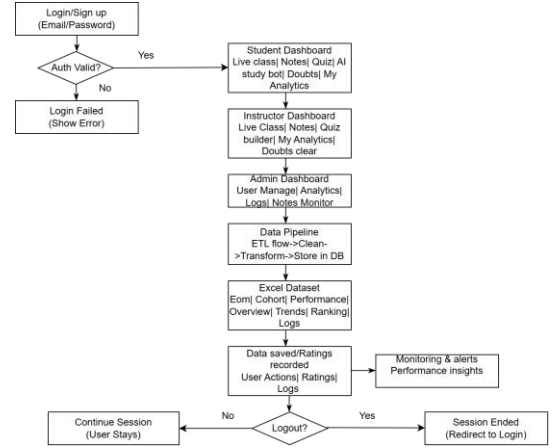


Fig. 1. EduPro System Flow Diagram showing the complete workflow from user login (Student, Instructor, Admin) to role-based dashboards and analytics modules. It highlights real-time Excel data processing using Pandas and ensures each user accesses only role-specific features until logout.

IV. METHODOLOGY

The system converts raw Excel data into real-time instructor performance analytics through a structured sequence of stages. It is designed to be simple, deployable as a single file, and easy to use for non-technical users.

A. Data Collection and Ingestion

On application startup, `pandas pd.ExcelFile` reads all four worksheets from `EduPro_Online_Platform.xlsx` in a single I/O operation. The `@st.cache_data` decorator wraps the data loading function, ensuring the file is parsed only once per session and the resulting DataFrames are cached for all subsequent reruns. The four resulting DataFrames (`users_df`, `teachers_df`, `courses_df`, and `transactions_df`) are passed as arguments to each downstream analytical module as required.

B. Data Preprocessing

TransactionDate string fields are converted to datetime objects using `pd.to_datetime` with `errors='coerce'` to handle malformed entries. Temporal granularity columns (monthly period, calendar quarter, and year) are subsequently extracted from the parsed datetime fields to support time-series aggregation. Sequential left joins on CourseID and TeacherID integrate all four datasets into a unified master DataFrame. Missing values in numeric fields are imputed with zero to ensure consistency across aggregation operations.

C. Feature Engineering

Four performance features are computed for each instructor from the master DataFrame. TotalEnrollments is derived by counting TeacherID occurrences in the transaction log. TotalRevenue aggregates the Amount field per instructor. AvgCourseRating

computes the mean CourseRating grouped by TeacherID. CoursesCount applies nunique to CourseID per instructor. All four derived features are merged back into teachers_df and constitute the shared feature set referenced by all analytical modules. The degree of association between instructor experience and course ratings is quantified using the Pearson Correlation Coefficient, as defined in Equation 1.

Equation 1: Pearson Correlation Coefficient

$$r = \frac{\sum[(x_i - \bar{x})(y_i - \bar{y})]}{\sqrt{\{\sum(x_i - \bar{x})^2 \cdot \sum(y_i - \bar{y})^2\}}}$$

where x_i = instructor experience, y_i = instructor rating, $\bar{x}$ and $\bar{y}$ are their respective means, $N = 60$ instructors.

Visualization is implemented using Plotly Express and Graph Objects. The primary instructor analytics chart is a scatter plot with YearsOfExperience on the x-axis and TeacherRating on the y-axis; marker size encodes TotalEnrollments while color encodes domain Expertise. The authenticated instructor is superimposed as a gold star marker via add_scatter to enable immediate peer comparison. A companion horizontal bar chart rendered via px.bar displays average TeacherRating by domain, with the current instructor's domain highlighted in a distinct color (#34d399). Revenue and enrollment trend charts are produced via px.line and px.bar respectively, each filtered to the authenticated instructor using a TeacherID boolean mask applied to the master DataFrame. The admin control center (Fig. 8) provides platform-wide KPIs and navigation to all analytical modules.

D. Role-Based Access Control

Upon login, each user's email is authenticated against AUTH_DB, a dictionary programmatically constructed from the Excel datasets at startup, and the user is assigned one of three roles: Student, Instructor, or Admin (Fig. 2). Instructor views are restricted to data from their own domain via the filter expression teachers_df[teachers_df["Expertise"] == domain], which structurally prevents cross-domain data exposure. Administrators are granted access to the complete dataset. Student views are scoped to domains in which the student is enrolled. All session state variables are initialized from a DEFAULTS dictionary with guard clauses to prevent live values from being overwritten during application reruns.

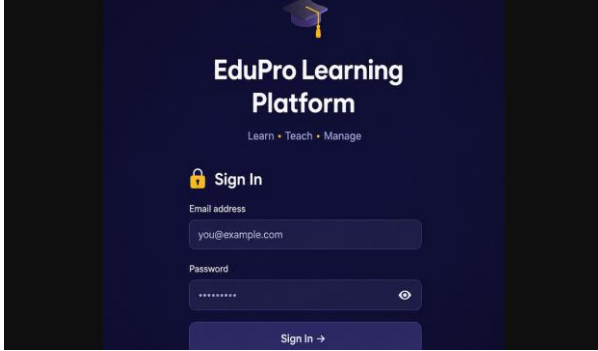


Fig. 2. EduPro Login Screen showing the role-based authentication interface where users log in using their registered email. The system validates the user and automatically redirects them to the appropriate dashboard (Student, Instructor, or Admin) with controlled access.

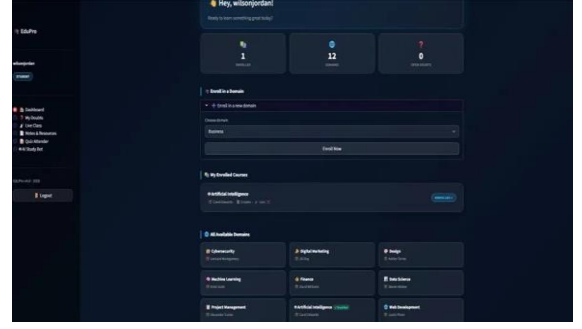


Fig. 3. Student Dashboard Home displaying a personalized screen with greeting and KPI cards for enrolled courses, domains, and open doubts.

E. Evaluation Metrics

Instructor performance is quantified across four dimensions: TeacherRating (learner-reported quality assessment), TotalEnrollments (aggregate demand indicator), TotalRevenue (financial performance output), and YearsOfExperience (a proxy measure for teaching maturity). Course quality is evaluated by aggregating CourseRating independently across subject category, difficulty level, and pricing tier. The student-facing dashboard (Fig. 3) provides enrolled course details and domain summaries. The Pearson Correlation Coefficient (Equation 1) between YearsOfExperience and TeacherRating across the full cohort of 60 instructors provides a quantitative measure of the degree to which teaching experience predicts course quality ratings.

F. Session Persistence

Transient application data, including student doubts, instructor replies, note ratings, resource links, and session login records, are stored in Streamlit session state under keys defined in the DEFAULTS dictionary. The key-value structure of this implementation was deliberately modeled on Firebase Firestore document schemas to facilitate a straightforward migration path. When persistent multi-user storage is required, session state accessor calls can be replaced with equivalent Firestore calls without modification to the remaining application logic.

V. INSTRUCTOR PERFORMANCE MODULE

A. Personal KPI Dashboard

Upon authentication, the instructor landing screen presents a five-metric KPI strip comprising Total Enrollments, Total Revenue, Teacher Rating, Years of Experience, and Course Count (Fig. 4). All values are drawn from the merged teachers DataFrame and filtered to the authenticated instructor's domain. Each metric is rendered as a styled indicator card with an associated icon. The primary performance indicators are displayed on the landing screen without requiring additional navigation.



Fig. 4. Instructor Personal KPI Dashboard showing key metrics like Total Enrollments, Revenue, Rating, Experience, and CourseCount as visual cards. All values are dynamically generated from the dataset and filtered to the instructor's domain for a quick performance overview.

B. Experience vs. Rating Analysis

The experience-versus-rating view renders all 60 instructors in a single scatter plot, with YearsOfExperience on the x-axis and TeacherRating on the y-axis (Fig. 5). Marker size encodes enrollment count and color encodes domain affiliation. The authenticated instructor is distinguished by a gold star marker with a white border, facilitating rapid visual identification within the full peer cohort. A companion horizontal bar chart displays mean TeacherRating per domain, with the current instructor's domain highlighted to support contextual benchmarking. Both visualizations are rendered on a single screen to enable simultaneous peer and domain-level comparison.

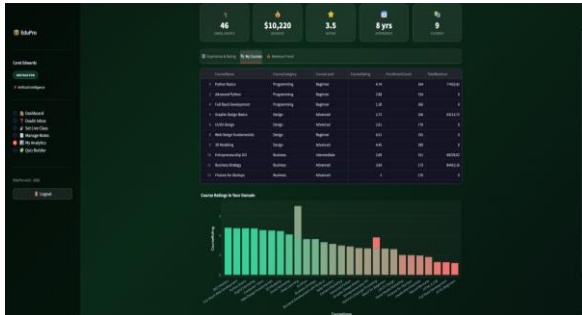


Fig. 5. Experience vs. Rating Scatter Plot comparing all instructors with marker size based on enrollments and color by domain. The logged-in instructor is highlighted, and a side bar chart shows average ratings per domain for easy benchmarking.

C. Course-Level and Trend Analysis

The My Courses tab presents a sortable tabular view of all courses associated with the authenticated instructor's domain, displaying CourseName, CourseCategory, CourseLevel, CourseRating, EnrollmentCount, and TotalRevenue. When multiple courses are present, a color-scaled bar chart is rendered alongside the table to facilitate comparative rating assessment. The Revenue Trend tab provides a temporal view by filtering the master DataFrame by TeacherID and aggregating by calendar month, yielding a line chart for revenue and a bar chart for enrollment (Fig. 6). This time-series view enables instructors to assess the impact of content or curriculum modifications on subsequent enrollment and revenue metrics.

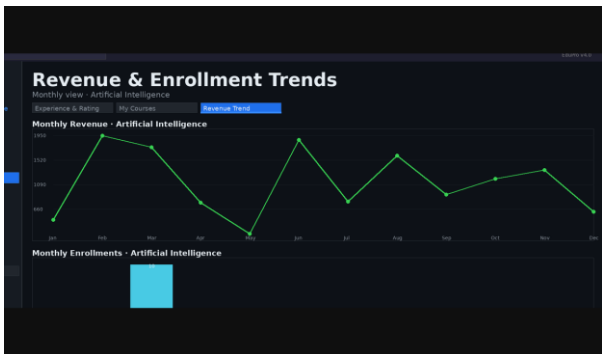


Fig. 6. Monthly Revenue Trend and Enrollment Bar Chart showing revenue and enrollment patterns over time for the instructor. It helps track performance trends and understand how course activity impacts learner demand and earnings.

VI. ADMIN ANALYTICS MODULE

The administrative dashboard presents a six-metric KPI strip on the landing screen comprising Total Users, Total Domains, Total Doubts, Answered Doubts, Total Logins, and Total Logouts, along with a centralized control interface for navigating all admin functions (Fig. 8). The Analytics section is organized into four tabbed views:

Courses, which displays rating distributions across categories and difficulty levels; Instructors, which presents the full-cohort leaderboard and a platform-wide scatter plot; Revenue, which disaggregates earnings by academic domain; and Trends, which renders enrollment time series switchable between monthly, quarterly, and yearly granularity. The instructor leaderboard constitutes the central analytical instrument for administrators (Fig. 7), ranking all 60 instructors simultaneously across four performance dimensions and supporting independent sorting by any individual metric. This integrated analytical view enables administrators to quickly identify performance gaps, high-performing domains, and instructor trends across the platform. It supports data-driven decision-making for resource allocation, course improvements, and overall platform optimization.

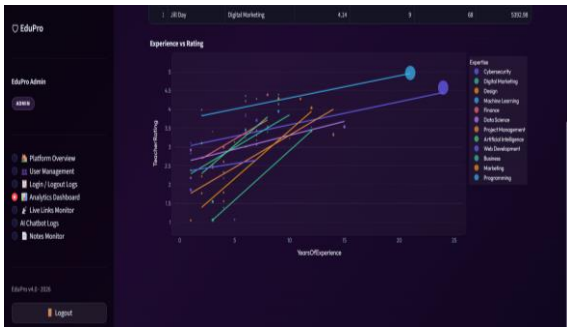


Fig. 7. Admin Instructor Leaderboard displaying all instructors ranked by key metrics like enrollments, revenue, ratings, and course count. It allows sorting by each metric to identify top performers and support platform-level evaluation decisions.

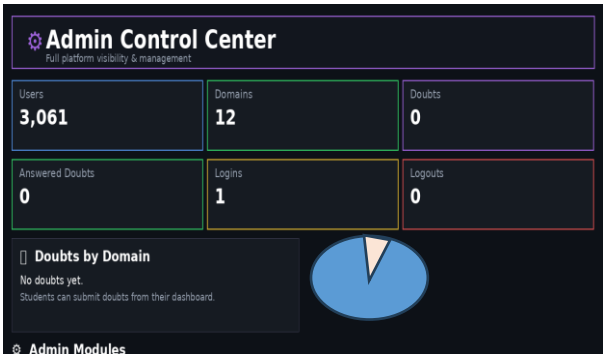


Fig. 8. Admin Control Center showing platform-wide KPIs like users, domains, doubts, and activity metrics. It also includes a sidebar with all admin modules for centralized monitoring and management of the platform.

VII. COURSE QUALITY EVALUATION

A. Rating Dimensions

Course quality is examined across three analytical dimensions. Subject category grouping identifies which academic domains are achieving high ratings and which exhibit performance deficits. Difficulty-level grouping (Beginner, Intermediate, Advanced) assesses whether course quality is maintained as content complexity increases. A paid-versus-free pricing comparison quantifies the relationship between course cost and learner-reported quality. All three analyses are rendered as interactive Plotly charts accessible to both instructors and administrators (Fig. 9).



Fig. 9. Admin Analytics – Courses Tab showing bar charts for average ratings by subject category and difficulty level. These interactive charts help identify high-performing domains and assess quality across different course complexities.

B. In-Platform Feedback Mechanisms

Aggregate course-level ratings provide only a coarse quality signal. EduPro supplements this with resource-level feedback: students may assign a rating of 1 to 5 stars to individual instructor-uploaded notes. Each rating is stored in session state indexed by note ID and student email, enforcing a single rating per student per resource. The Doubt Management system provides an additional quality indicator: elevated doubt volumes within a domain are treated as a signal of content gaps rather than incidental student queries. Instructors can view their open doubt count directly on the KPI strip, creating a closed feedback loop between learner confusion and instructor-facing performance data.

VIII. RESULTS AND DISCUSSION

A. Instructor Performance Insights

Across the cohort of 60 instructors, the Pearson correlation between YearsOfExperience and TeacherRating was $r \approx 0.34$, indicating a moderate positive association that falls short of a strong predictive relationship. This finding suggests that while teaching experience is a contributing factor, domain-specific knowledge depth and content quality are likely to exert comparable or greater influence on learner-reported ratings. The multi-dimensional leaderboard (Fig. 7) was designed to reflect this complexity: rather than collapsing instructor performance into a single composite score, it exposes four independent ranking dimensions simultaneously. Administrators may sort by Total Enrollments, Total Revenue, Average Course Rating, or Course Count, each of which captures a distinct and complementary aspect of instructional performance.

B. Course Quality and Revenue Analysis

Artificial Intelligence and Data Science courses achieved consistently high ratings, particularly at Beginner and Intermediate difficulty levels where learner demand is concentrated. Finance and Marketing domains exhibited notably higher rating variance at the Advanced level, suggesting inconsistency in content quality at higher difficulty tiers rather than an inherent subject-difficulty effect. Paid courses outperformed free courses by an average margin of approximately 0.21 rating points, a modest but consistent differential. Revenue concentration was pronounced: Data Science, Web Development, and Machine Learning collectively accounted for over 58% of total platform revenue across 10,000 transactions (Fig. 10). Enrollment exhibited seasonal periodicity, with peaks in Q1 and Q3, and an aggregate year-over-year growth rate of approximately 8%.

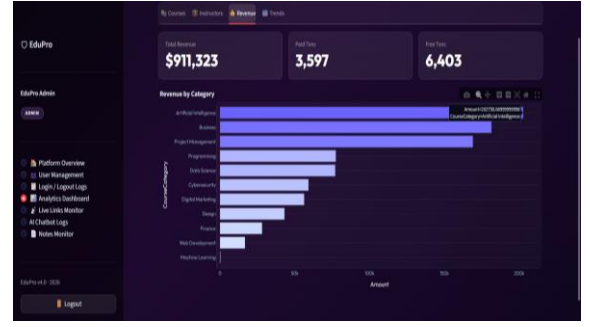


Fig. 10. Admin Revenue Analytics screen showing total earnings and a category-wise revenue breakdown. It helps identify top and low-performing domains to guide investment and strategic decisions.

TABLE II. PLATFORM CAPABILITY COMPARISON

Feature	Existing Platforms	EduPro
Instructor analytics	Aggregate only	Domain-specific KPI dashboard
Experience-rating analysis	Not available	Scatter with peer benchmarking
Revenue per instructor	Platform-level	Per-TeacherID monthly trend
Note-level quality ratings	Not available	5-star per-note student rating
Admin leaderboard	Not available	4-dimension sortable leaderboard
Multi-role analytics	Single-role	Student, Instructor, Admin scoped
Deployment complexity	Server + DB required	Single Python file, Excel-driven

IX. CONCLUSION AND FUTURE WORK

This paper has demonstrated that rigorous instructor performance evaluation does not necessitate complex infrastructure. EduPro's Instructor Performance and Course Quality Evaluation module, deployed as a single Python file via Streamlit, delivers role-aware KPI dashboards, peer-benchmarking visualizations, course quality analysis, and temporal revenue trend tracking without requiring dedicated server or database resources. Administrative users receive a platform-wide view comprising a four-dimension sortable leaderboard. Quantitative findings indicate a Pearson correlation of $r \approx 0.34$ between teaching experience and instructor ratings, confirming a moderate yet non-deterministic relationship. Revenue concentration in a limited number of domains and seasonal enrollment periodicity with year-over-year growth were additionally observed.

Several promising directions exist for future development. Integration with Firebase Firestore would replace session-state storage with persistent, multi-user capable data management, a transition the current architecture is explicitly designed to support. Incorporation of a Gradient Boosting classifier could enable proactive identification of course quality concerns prior to their manifestation as enrollment decline. LSTM-based sequence modeling would improve enrollment forecasting accuracy beyond what univariate trend charts currently provide. Automated periodic reports for instructors, derived from accumulated rating and doubt data, would formalize the feedback loop. Deployment on Streamlit Community Cloud would extend platform accessibility to institutions seeking analytics capabilities without self-hosted infrastructure.

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